

SET perihelion derivation 16

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We determine the stationary spherical burden and radial response from SET's source-surface relation, physical work-current law and consistency across a common family of sources. We then derive the orbit from the capacity action. The normalized source readout enters an exact equivalent energy equation, which connects the receiving clock's normalization to the orbital calculation. The resulting leading advance is $6\pi GM/[a(1-e^2)c^2]$, approximately 42.98 arcseconds per century for Mercury.

SET takes continuous sourcing of field burden, or occupied capacity, as a foundational physical principle. The source law specifies the source contribution, while the capacity relation describes its participation in local physical processes. The stationary field construction follows derivation 14 [1], with physical-current balance and a universal response class stated explicitly. Here we explain the probe action through calibrated event progression, develop the connection to normalized burden and show the intervening motion equations. The field is determined before the orbital correction is calculated.

The calculation concerns an isolated, nonrotating, positive spherical source and a test body in its stationary vacuum exterior. The reference time is normalized at infinity, and the held-clock capacity is positive throughout the region considered. The source interior, other bodies and radiation reaction are outside this calculation.

1 Capacity and the radial motion response

Begin with the SET capacity relation [2]:

$$S^2 + V_{\text{rel}}^2 + c^2 \beta^2 = c^2, \beta = \frac{d\tau}{dT}. \quad (1)$$

Here S is gravitational burden in speed units, T is the stationary reference time, and τ records the probe's clock progression. For a held observer, define

$$A = \alpha^2 = 1 - \frac{S^2}{c^2}, d\tau_h = \alpha dT. \quad (2)$$

Use areal radius r , so a spherical screen has area $A_\Sigma = 4\pi r^2$. The dimensionless A and the area A_Σ are distinct. In the equatorial plane, write

$$V_{\text{rel}}^2 = \zeta^2 \dot{r}^2 + r^2 \dot{\phi}^2, d\ell = \zeta |dr|. \quad (3)$$

Dots denote d/dT . The positive factor ζ converts an areal-radius displacement to the radial distance entering the local motion measurement. It will be determined by the field construction rather than fixed at the outset.

2 The source surface and the response class

Let the source have mass parameter M and physical surface radius R . The source law uses the spherical reference volume and the surface burden:

$$V_{\text{ref}} = \frac{4\pi R^3}{3}, Q_{\Omega}^2 = 24\pi GM V_{\text{ref}} = 32\pi^2 GM R^3. \quad (4)$$

$$Q_{\Omega} = 4\pi R^2 S(R). \quad (5)$$

Q_{Ω} is assigned to this source domain. M denotes the total source mass-energy parameter used consistently in the boundary and exterior equations. The reference volume is the volume specified by the source law; it is not replaced by an integral over an as-yet unknown physical radial measure.

For the positive source branch, equation (4) is $Q_{\Omega} = \sqrt{32\pi^2 GM R^3} = 4\pi \sqrt{2GM R^3}$. Its squared form and the ratio $Q_{\Omega}^2 / (24\pi V_{\text{ref}}) = GM$ are algebraic expressions of the same law. The surface relation places that source law in the capacity budget through $S(R) = Q_{\Omega} / (4\pi R^2)$. A change to receiving-clock units, treated in Section 6, is a separate conversion of the readout.

Squaring equation (5) and comparing with equation (4) gives

$$16\pi^2 R^4 S(R)^2 = 32\pi^2 GM R^3, S(R)^2 = \frac{2GM}{R}. \quad (6)$$

$$A_R = 1 - \frac{R_s}{R}, R_s = \frac{2GM}{c^2}, R > R_s. \quad (7)$$

This fixes the burden at an actual source surface. We next determine how the same source establishes the exterior response. Parameterize the radial factor by

$$\zeta^2 = \frac{H(A)}{A}, H(A) > 0, H(1) = 1. \quad (8)$$

We take H to be positive, continuous and universal across the specified stationary spherical source family, with the differentiability required for the motion equations. Its only independent argument in this class is the local capacity A . The class excludes separate dependence on source size, density gradients, history or an additional field. This restriction is part of the proposed spherical response law; spherical symmetry alone does not impose it.

The unloaded limit has $A \rightarrow 1$ and $\zeta \rightarrow 1$ at infinity. The allowed source family is assumed to span a continuous interval of surface capacities extending to this limit. Section 4 uses that common family to determine H ; it does not move the boundary of one fixed source through vacuum.

3 Motion from calibrated event progression

The capacity relation assigns a clock-progression factor β to each proposed probe path. For a stable clock, calibrated event accumulation along that path is proportional to $\int \beta dT$. We use this accumulation to define the action of a structureless test body with constant rest mass and no additional direct field coupling.

The motion principle is stationarity under path variations with fixed endpoint events. With a positive constant κ carrying energy units, write

$$I_p = -\kappa \int \beta dT, \delta I_p = 0. \quad (9)$$

Each interval contributes the same action per unit of calibrated event progression. This is the minimal coupling adopted here. The invariant determines β ; stationarity determines which path the free probe follows. It does not require every physical path to maximize an event count.

Calibrate the coefficient with the probe's measured inertial mass. In the unloaded limit $S=0$ and $\zeta=1$, write $V_{\text{rel}}=v$ and expand at low speed:

$$L_p = -\kappa \sqrt{1 - \frac{v^2}{c^2}} = -\kappa + \frac{\kappa v^2}{2c^2} + O\left(\frac{\kappa v^4}{c^4}\right). \quad (10)$$

The kinetic term must be $m_p v^2 / 2$. Therefore

$$\frac{\kappa}{c^2} = m_p, \kappa = m_p c^2. \quad (11)$$

The calibrated probe action is consequently

$$I_p = -m_p c^2 \int \beta dT, L_p = -m_p c^2 \beta. \quad (12)$$

This fixes the coefficient without using an orbital precession. It also gives the gravitational coupling directly. For $S^2/c^2 \ll 1$ and $V_{\text{rel}}^2/c^2 \ll 1$, expand the same action:

$$L_p = -m_p c^2 + \frac{m_p}{2} (V_{\text{rel}}^2 + S^2) + O\left(\frac{m_p (S^2 + V_{\text{rel}}^2)^2}{c^2}\right). \quad (13)$$

At leading order in the unloaded spatial measure, the specific potential and the acceleration are

$$\Phi = -\frac{S^2}{2}, a = -\nabla \Phi = \frac{1}{2} \nabla S^2. \quad (14)$$

The attractive response follows because the spherical burden decreases outward. The higher-order orbital terms require the full action and the radial response, which we now determine together with the exterior field.

Physical work and the exterior field law

Combining equations (1) and (3) gives $\beta^2 = A - (\zeta^2 \dot{r}^2 + r^2 \dot{\phi}^2) / c^2$. Differentiation of the action yields

$$p_r = \frac{m_p \zeta^2 \dot{r}}{\beta}, p_\phi = \frac{m_p r^2 \dot{\phi}}{\beta}. \quad (15)$$

$$E = p_r \dot{r} + p_\phi \dot{\phi} - L_p = \frac{m_p c^2 A}{\beta}. \quad (16)$$

The final equality uses $\zeta^2 \dot{r}^2 + r^2 \dot{\phi}^2 = c^2 (A - \beta^2)$. Locally, the motion factor is $\gamma = \alpha / \beta$, so $E = \alpha m_p c^2 \gamma$. A held probe therefore has $E_{\text{hold}} = m_p c^2 \alpha$.

For a quasistatic outward lift, local work $F_{\text{hold}} d\ell$ contributes $\alpha F_{\text{hold}} d\ell$ to the reference energy. Equating this to dE_{hold} gives

$$\alpha F_{\text{hold}} \zeta dr = m_p c^2 \frac{d\alpha}{dr} dr. \quad (17)$$

$$a_{\text{hold}} = \frac{c^2}{\zeta} \frac{d \ln \alpha}{dr}, J \equiv \alpha a_{\text{hold}} = \frac{c^2}{2\sqrt{H(A)}} \frac{dA}{dr}. \quad (18)$$

J is the outward redshifted support acceleration. Defining the radial increment $d\chi = \alpha d\ell = \sqrt{H} dr$ for outward displacements also gives $J = -\frac{1}{2} dS^2 / d\chi$. The quantity has units of acceleration; it is not the volume-per-time readout Q .

We retain the spherical physical-current law of the earlier SET construction [1, 3]: every complete vacuum screen carries the same source charge in this work-current measure,

$$\oint_{\Sigma_r} J \cdot dA_\Sigma = 4\pi GM, r^2 J = GM. \quad (19)$$

Equation (19) states the spherical field-response principle: the work-current carries the source charge through the vacuum exterior. Its value is fixed by the source law through $GM = Q_\Omega^2 / (24\pi V_{\text{ref}})$. This identifies the physical gradient to which conservation applies. Together with the universal response class, it defines the field problem whose solution follows next.

4 Source consistency determines the radial response

Combine the work-current expression with the complete-screen law:

$$\frac{r^2 c^2}{2\sqrt{H(A)}} \frac{dA}{dr} = GM. \quad (20)$$

Separate variables and integrate from the actual surface R to the unloaded exterior limit:

$$\frac{dA}{\sqrt{H(A)}} = \frac{R_s}{r^2} dr, \quad (21)$$

$$\int_{A_R}^1 \frac{du}{\sqrt{H(u)}} = R_s \int_R^\infty \frac{dr}{r^2} = \frac{R_s}{R}. \quad (22)$$

The independently retained source-surface relation gives $R_s/R = 1 - A_R$. Thus agreement of the source and field requires

$$\int_{A_R}^1 \frac{du}{\sqrt{H(u)}} = 1 - A_R. \quad (23)$$

Require this identity for the continuous family of source surfaces, using the same function H . Differentiating with respect to the family's boundary capacity gives

$$-\frac{1}{\sqrt{H(A_R)}} = -1, H(A_R) = 1. \quad (24)$$

The fundamental theorem of calculus applies because H is positive and continuous. If the admissible family covers $A_{min} \leq A_R < 1$, it determines the response throughout that interval:

$$H(A) = 1, \zeta^2 = \frac{1}{A}. \quad (25)$$

Each member of the family has its own actual source radius, reference volume and source readout. The differentiation compares those physical configurations. It does not equate a fixed Q_Ω to every exterior sphere's area times its local burden.

The conclusion applies to the exterior capacities covered by the family. Extending it below A_{min} would require a corresponding extension of the admissible source family. This argument does not establish the stability of arbitrarily compact sources. Appendix A shows explicitly why agreement at only one source surface would not determine the entire function.

5 The burden profile and the exterior source reading

With $H = 1$, the field equation has become

$$\frac{dA}{dr} = \frac{2GM}{c^2 r^2}, A(\infty) = 1. \quad (26)$$

Integrating from r to infinity gives the local profile:

$$1 - A(r) = \int_r^\infty \frac{2GM}{c^2 u^2} du = \frac{2GM}{r c^2}, \quad (27)$$

$$A(r) = 1 - \frac{2GM}{r c^2}, S(r)^2 = \frac{2GM}{r}. \quad (28)$$

The same calculation supplies the inverse-area work-current and the burden-gradient relation:

$$J = \frac{GM}{r^2} = -\frac{1}{2} \frac{dS^2}{dr}, -\frac{r^2}{2} \frac{dS^2}{dr} = GM. \quad (29)$$

Before selection, the physical gradient contained the factor $1/\sqrt{H}$. Both that radial response and the profile are now fixed. The older carrier relation follows as an algebraic consequence:

$$S^4 = \frac{4G^2 M^2}{r^2}, J = \frac{S^4}{4GM}. \quad (30)$$

Define the local spherical screen readout, distinguishing it from the fixed source-domain quantity:

$$Q_0(r) = 4\pi r^2 S(r) = 4\pi \sqrt{2GM} r^3. \quad (31)$$

$$\frac{Q_0(r)^2}{24\pi V_{\text{ref}}(r)} = \frac{rS(r)^2}{2} = GM, V_{\text{ref}}(r) = \frac{4\pi r^3}{3}. \quad (32)$$

The source charge read this way is independent of the exterior screen. This constancy is a result of the solved field. $Q_0(r)$ itself grows as $r^{3/2}$ and is not a conserved volume discharge across successive spheres. At the material surface, $Q_0(R) = Q_\Omega$.

6 The receiving clock and normalized burden

The same screen readout can be stated per held-clock second. Keeping its spatial normalization unchanged, define

$$Q_* d\tau_h = Q_0 dT, Q_* = \frac{Q_0}{\alpha}. \quad (33)$$

The solved exterior then gives the normalized expression used in the earlier SET algebra:

$$Q_*(r) = 4\pi r^{3/2} \sqrt{\frac{2GM}{1 - 2GM/(rc^2)}}. \quad (34)$$

The physical source charge has not changed. A slower receiving clock assigns a larger rate to the same specified readout. If a starting rate were defined per source proper time instead of per T , its conversion would also include the source clock factor.

Introduce a speed-valued normalized burden, distinguished from the local capacity burden S :

$$s_* \equiv \frac{Q_*}{A_\Sigma} = \frac{S}{\alpha}. \quad (35)$$

The held capacity relation alone gives the following exact identities:

$$\frac{s_*^2}{c^2} = \frac{1 - A}{A}, A = \frac{1}{1 + s_*^2/c^2}. \quad (36)$$

$$S^2 = \frac{s_*^2}{1 + s_*^2/c^2}. \quad (37)$$

In the solved spherical profile, this becomes

$$s_*^2 = \frac{2GM/r}{1 - R_s/r} = \frac{2GM}{r - R_s}. \quad (38)$$

This expression is not a new areal radius or a replacement for S in the invariant. Its denominator comes from the receiving clock's normalization. Section 8 derives its role in an equivalent orbital energy equation rather than treating it as an additional force.

For a moving receiving clock the same rate bookkeeping uses $Q_{\text{rec}} = Q_0/\beta$. The factors α and β coincide only for a held observer. No literal transport or Doppler interception law is inferred solely by this change of clock units.

7 The same response in propagation and escape

In the selected exterior, the radial motion coefficient has a compact expression in normalized burden:

$$\zeta^2 = \frac{1}{A} = 1 + \frac{s_\star^2}{c^2}. \quad (39)$$

Its equality to $1/A$ follows from Section 4. The identities of Section 6 express that selected response through $s_\star$; they do not select it a second time.

Use the retained propagation coupling in which a radial light signal has zero clock progression in the same interval. Setting $\beta = 0$ in the radial capacity equation yields

$$\left| \frac{dr}{dT} \right| = \frac{c\alpha}{\zeta} = cA = \frac{c}{1 + s_\star^2/c^2}. \quad (40)$$

$$\frac{d\ell}{dT} = c\alpha, \frac{d\ell}{d\tau_h} = c. \quad (41)$$

The local clock and the signal propagation participate in the same response. For two nearby radial locations, the reference-clock round-trip time is

$$dT_{\text{round}} = \frac{2|dr|}{cA} = \frac{2|dr|}{c} \left(1 + \frac{s_\star^2}{c^2} \right). \quad (42)$$

The local radar distance obtained with the emitter's held clock is $c\alpha dT_{\text{round}}/2 = |dr|/\alpha$. A finite path requires integration through its varying capacity; the clock at a fixed emitter multiplies the complete reference-time interval by that emitter's lapse.

The distinction between S and $s_\star$ is also fixed by the escape calculation. The probe energy from Section 3 can be written

$$\varepsilon \equiv \frac{E}{m_p c^2} = \frac{A}{\beta} = \alpha\gamma, \gamma = \frac{\alpha}{\beta}. \quad (43)$$

Marginal escape to rest at infinity has $\varepsilon = 1$. For the local speed measured by a held observer, this gives

$$\gamma_{\text{esc}} = \frac{1}{\alpha}, v_{\text{esc}}^2 = c^2(1 - \alpha^2) = S^2. \quad (44)$$

Thus S remains the local marginal escape speed in this branch. The normalized quantity is $s_\star = \gamma_{\text{esc}} v_{\text{esc}}$; it is not another local light-speed limit. Appendix D gives the optional free-fall representation after the response has been selected.

8 An exact orbital energy equation from normalized burden

We now derive the normalized-burden form of the motion equations. Temporarily leave $\zeta(r)$ explicit so that the transformation can be checked without presupposing the selected response. The action has no explicit dependence on T or φ . Its constants are

$$\varepsilon = \frac{A}{\beta}, j = \frac{r^2 \dot{\varphi}}{\beta} = r^2 \frac{d\varphi}{d\tau}. \quad (45)$$

The parameter j is angular momentum per unit probe mass. Introduce an auxiliary parameter along this fixed-energy orbit:

$$d\sigma = \varepsilon d\tau = A dT, \ell_\sigma = \frac{j}{\varepsilon} = r^2 \frac{d\varphi}{d\sigma}. \quad (46)$$

σ is a computational parameter. It is neither the reference clock T , the held clock τ_h nor the probe clock τ . Since ε is constant, it is a constant rescaling of the probe's proper time along this orbit.

Divide the capacity equation by A^2 and use $\beta = A / \varepsilon$:

$$\frac{1}{\varepsilon^2} = \frac{1}{A} - \frac{1}{c^2} \left[\zeta^2 \left(\frac{dr}{d\sigma} \right)^2 + r^2 \left(\frac{d\varphi}{d\sigma} \right)^2 \right]. \quad (47)$$

Now use $s_\star^2 = c^2 (1/A - 1)$ and rearrange:

$$\frac{1}{2} \left[\zeta^2 \left(\frac{dr}{d\sigma} \right)^2 + r^2 \left(\frac{d\varphi}{d\sigma} \right)^2 \right] - \frac{s_\star^2}{2} = E_\sigma, \quad (48)$$

$$E_\sigma = \frac{c^2}{2} (1 - \varepsilon^{-2}), \Phi_\sigma = -\frac{s_\star^2}{2}. \quad (49)$$

E_σ is an auxiliary conserved energy per mass; physical reference energy remains $E = m_p c^2 \varepsilon$. The original radial coefficient is still present. In the selected spherical field, the effective potential is

$$\Phi_\sigma(r) = -\frac{GM}{r - R_s}. \quad (50)$$

We have obtained this potential variable by transforming the retained action's conserved quantities. To establish full equivalence, the next step verifies its motion equation, not only its energy integral.

Equivalence of the motion equations

For compact notation on this page, write $q = \zeta^2$, $r_\tau = dr/d\tau$ and $r_{\tau\tau} = d^2r/d\tau^2$; a subscript r denotes a field derivative. Variation of L_p gives

$$\frac{d}{dT} \left(\frac{q \dot{r}}{\beta} \right) = -\frac{c^2 A_r}{2\beta} + \frac{q_r \dot{r}^2}{2\beta} + \frac{r \dot{\phi}^2}{\beta}. \quad (51)$$

Use $d/dT = \beta d/dr$ and $\dot{r} = \beta r_\tau$. Divide by β , expand the derivative on the left, and move one half of the q_r term to that side:

$$q r_{\tau\tau} + \frac{q_r}{2} r_\tau^2 - r \phi_\tau^2 = -\frac{c^2 A_r}{2\beta^2} = -\frac{c^2 \varepsilon^2 A_r}{2A^2}. \quad (52)$$

Since $d\sigma = \varepsilon d\tau$ and ε is constant, each quadratic velocity or second derivative contributes the same factor ε^2 . After division,

$$q r_{\sigma\sigma} + \frac{q_r}{2} r_\sigma^2 - r \phi_\sigma^2 = -\frac{c^2 A_r}{2A^2}. \quad (53)$$

Differentiate the normalized-burden identity:

$$\frac{1}{2} \frac{ds_\star^2}{dr} = \frac{c^2}{2} \frac{d}{dr} \left(\frac{1}{A} - 1 \right) = -\frac{c^2 A_r}{2A^2}. \quad (54)$$

Equations (53) and (54), together with $r^2 \phi_\sigma = \ell_\sigma$, are the equations of the auxiliary Lagrangian

$$\frac{L_\sigma}{m_p} = \frac{1}{2} (q r_\sigma^2 + r^2 \phi_\sigma^2) + \frac{s_\star^2}{2}. \quad (55)$$

This establishes equivalence of both energy and motion for each orbit. We transformed the original Euler–Lagrange equation before identifying L_σ ; a trajectory-dependent substitution of time directly into an action need not preserve its variational problem.

Normalized burden therefore has an exact role as the potential variable for these parameterized trajectories. Its radial derivative and the radial motion coefficient must be used together. Adding a second enhancement to the force because $Q_\star > Q_0$ would count the same clock normalization twice.

9 The exact exterior orbit equation

Use the selected response $q = 1/A$ in equation (48). Put $w = 1/r$ and let a prime on w denote $d/d\varphi$. Since $r_\sigma = -\ell_\sigma w'$, the energy equation becomes

$$\frac{\ell_\sigma^2}{A} (w')^2 + \ell_\sigma^2 w^2 = s_\star^2 + c^2 (1 - \varepsilon^{-2}). \quad (56)$$

Multiply by A and use $A s_\star^2 = c^2 (1 - A)$:

$$\ell_\sigma^2 (w')^2 + A \ell_\sigma^2 w^2 = c^2 (1 - A \varepsilon^{-2}). \quad (57)$$

Finally multiply by ε^2 , recalling $j = \varepsilon \ell_\sigma$:

$$j^2 (w')^2 = c^2 \varepsilon^2 - A (c^2 + j^2 w^2). \quad (58)$$

This is also obtained directly from the original capacity action using $dT/d\tau = \varepsilon/A$ and $dr/d\tau = -j w'$. The normalized-burden and original-clock routes therefore agree before any weak-field expansion.

Substitute the solved $A = 1 - 2GMw/c^2$ and expand the product:

$$j^2 (w')^2 = c^2 (\varepsilon^2 - 1) + 2GMw - j^2 w^2 + \frac{2GM j^2 w^3}{c^2}. \quad (59)$$

Differentiate with respect to φ :

$$2j^2 w' w'' = 2GM w' - 2j^2 w w' + \frac{6GM j^2 w^2 w'}{c^2}. \quad (60)$$

Where $w' \neq 0$, divide by $2j^2 w'$. The smooth second-order equation extends to isolated turning points by continuity:

$$w'' + w = \frac{GM}{j^2} + \frac{3GM w^2}{c^2}. \quad (61)$$

The coefficient 3 comes from differentiating the cubic term generated by the solved response and the angular-motion term. It has not been adjusted to a measured advance. Circular motion follows directly from the radial Euler–Lagrange equation and does not require division by w' .

10 From the orbit equation to the perihelion advance

Define $\ell_g = GM / c^2$ and $p_0 = j^2 / (GM)$. The unperturbed solution is

$$w_0 = \frac{1 + e \cos \varphi}{p_0}, \quad w_0'' + w_0 = \frac{1}{p_0}. \quad (62)$$

Write $w = w_0 + \delta w$ and retain first order in ℓ_g / p_0 . In the correction term, replace w by w_0 :

$$\delta w'' + \delta w = 3 \ell_g w_0^2. \quad (63)$$

$$3 \ell_g w_0^2 = \frac{3 \ell_g}{p_0^2} \left(1 + \frac{e^2}{2} + 2 e \cos \varphi + \frac{e^2}{2} \cos 2 \varphi \right). \quad (64)$$

The constant and double-frequency terms give bounded changes. The term proportional to $\cos \varphi$ is resonant with the unperturbed radial oscillation. Use

$$\left(\frac{d^2}{d\varphi^2} + 1 \right) (\varphi \sin \varphi) = 2 \cos \varphi, \quad (65)$$

to obtain a particular correction:

$$\delta w = \frac{3 \ell_g}{p_0^2} \left(1 + \frac{e^2}{2} + e \varphi \sin \varphi - \frac{e^2}{6} \cos 2 \varphi \right). \quad (66)$$

The homogeneous terms shift the fitted eccentricity and initial phase. To read the accumulated advance, compare the resonant term with

$$\frac{e}{p_0} \cos[(1 - \delta) \varphi] \simeq \frac{e}{p_0} \cos \varphi + \frac{e \delta}{p_0} \varphi \sin \varphi. \quad (67)$$

$$\delta = \frac{3 \ell_g}{p_0}, \quad \Delta \varphi = \frac{2 \pi}{1 - \delta} - 2 \pi \simeq \frac{6 \pi \ell_g}{p_0}. \quad (68)$$

To this order $p_0 = a(1 - e^2)$, giving the advance per radial period:

$$\Delta \varphi = \frac{6 \pi GM}{a(1 - e^2)c^2} + O\left(\frac{G^2 M^2}{a^2(1 - e^2)^2 c^4}\right). \quad (69)$$

The final formula assumes a weak-field bound orbit. Equation (61) itself is exact within the selected stationary spherical test-body model.

11 Exact orbit verification and the Mercury scale

For an exact check, specify the turning radii by $r_p = p/(1+e)$ and $r_a = p/(1-e)$, with $P = p/\ell_g$. Setting equation (58) to zero at each turning point and subtracting the two equations gives

$$\frac{j^2}{GM \ell_g} = \frac{P^2}{P-3-e^2}, \epsilon^2 = \frac{(P-2)^2 - 4e^2}{P(P-3-e^2)}. \quad (70)$$

Here p is defined by the exact turning radii. It differs from $p_0 = j^2/(GM)$ beyond leading order. Substitute $w = (1 + e \cos \xi)/p$ into equation (59):

$$(w')^2 = \frac{e^2 \sin^2 \xi}{p^2} \left[1 - \frac{2(3 + e \cos \xi)}{P} \right]. \quad (71)$$

For $P > 6 + 2e$, cancellation against $(dw/d\xi)^2$ gives the total-angle integral and advance:

$$\Delta\phi = \int_0^{2\pi} \left[\left(1 - \frac{2(3 + e \cos \xi)}{P} \right)^{-1/2} - 1 \right] d\xi. \quad (72)$$

Independent numerical quadrature and direct integration of equation (61) agree. The values below use $e = 0.2$ and express p in units of GM/c^2 . They are checks of the equations, not additional astronomical observations.

P	Advance from quadrature rad	Advance from orbit integration rad
100	0.1974475160	0.1974475160
1000	0.0189349966	0.0189349966
10000	0.0018858061	0.0018858061

The same calculation with the alternative radial response is carried through in Appendix B. The normalized energy and motion transformation was also checked with exact rational arithmetic before the numerical integrations.

The Mercury benchmark

Use the solar gravitational parameter and constants from NASA JPL [4], together with JPL's approximate J2000 Mercury elements [5]:

$$G M_{\odot} = 1.3271244004127942 \times 10^{20} \text{ m}^3/\text{s}^2, \quad (73)$$

$$c = 299792458 \text{ m/s}, 1 \text{ au} = 149597870700 \text{ m}, \quad (74)$$

$$a = 0.38709927 \text{ au}, e = 0.20563593. \quad (75)$$

These give the orbital scale and expansion parameter

$$p = a(1 - e^2) = 5.546046913 \times 10^{10} \text{ m}, \quad (76)$$

$$\frac{G M_{\odot}}{p c^2} = 2.662482056 \times 10^{-8}. \quad (77)$$

Equation (69) then gives

$$\Delta \phi_{\text{orbit}} = 5.018660440 \times 10^{-7} \text{ rad} = 0.1035173023 \text{ arcsec}. \quad (78)$$

With 36525 days per Julian century [4] and the listed sidereal period 87.969 days [6],

$$\Delta \phi_{\text{century}} = 0.1035173023 \frac{36525}{87.969} \text{ arcsec} = 42.98070 \text{ arcsec}. \quad (79)$$

This is the standard approximately 43 arcseconds per century spherical relativistic contribution [7]. The source and response coefficients were fixed before this numerical substitution. The approximate elements are benchmark inputs, not a new fit to Mercury's observations; their precision does not support interpreting all displayed digits as empirical accuracy.

The calculation neglects the other planets, source rotation and multipoles, and test-body backreaction. Recovering the spherical contribution does not constitute a calculation of Mercury's complete observed precession. Appendix C uses the same inputs to evaluate the normalized readout during the finite time spent near perihelion.

12 What the spherical construction establishes

The field calculation requires the physical work-current to carry the source charge and to agree with the source-surface law across a universal family. This selects both $S^2 = 2GM/r$ and $\zeta^2 = 1/A$ on the family's covered interval. Stationary calibrated event progression supplies the probe action, with its coefficient fixed by inertial mass. That action gives the conserved orbital energy and the exact orbit. The normalized-burden representation is an equivalent form of the same motion.

The logical dependencies are explicit:

Quantity or relation	Status in this construction
Capacity relation	Retained SET capacity principle
Probe action	Stationary event progression with coefficient fixed by inertial mass
Source law and source-surface correspondence	Retained source principles
Physical-current balance and universal $H(A)$	Specified spherical field-response premises
Radial conversion and exterior burden	Determined by consistency and the exterior boundary condition
Normalized orbital potential	Derived by an exact change of orbital parameter
Perihelion equation and leading advance	Derived from the selected field and probe action

The solved clock and motion relations are mathematically equivalent to the Schwarzschild exterior. That equivalence is a result to check; an Einstein field equation or a metric-completion action did not select the response in this route. Appendix D obtains the familiar free-fall representation after the field has been solved.

Orbital energy and field energy answer different questions. Sections 3 and 8 derive the probe's conserved energy and its normalized-burden representation. A field-energy functional assigns energy to field configurations; an appropriate variational principle can then supply field equations. Here the stationary equations are stated directly through current balance and the response class. Selecting a separate field functional is not required to solve this exterior problem. Equivalent action expressions can yield the same equations; the relevant requirement is determinate dynamics.

The result is a complete conditional spherical derivation from the displayed SET principles. The current law and universal response class remain explicit field premises; they are not claimed to follow uniquely from the scalar capacity relation and finite-domain source law alone. Within those premises, the exterior, normalized orbital energy and perihelion follow without an additional field-energy ansatz or an extension to time-dependent field evolution.

Appendix A Why the source family is needed

Write $x = 1 - A_R$ and define the integrated response

$$F(x) = \int_0^x \frac{dt}{\sqrt{H(1-t)}}. \quad (80)$$

The source-field correspondence requires $F(x) = x$. At only one chosen boundary $x = x_0$, this is one integral constraint. As an explicit control, take

$$F(x) = x + \eta x^2 (x - x_0), x_0 = \frac{1}{2}, \eta = \frac{1}{5}. \quad (81)$$

$$F'(x) = 1 + \eta(3x^2 - 2x_0x), H(1-x) = \frac{1}{[F'(x)]^2}. \quad (82)$$

This derivative is positive on $0 \leq x \leq 1$. The response has $H(1) = 1$ and satisfies $F(x_0) = x_0$, yet

$$H(1-x_0) = \frac{1}{(1+\eta x_0^2)^2} = 0.90702948 \neq 1. \quad (83)$$

This is a mathematical control, not a proposed complete alternative theory. It shows why one boundary agreement does not select a function. Requiring $F(x) = x$ on a continuous interval instead gives $F'(x) = 1$ and hence $H = 1$ on that interval.

A second control follows an alternative through the actual field equation. Let $H(A) = 1 - b(1-A)^2$, with $b = 1/2$, and put $y = R_s/r$ and $x = 1 - A(r)$. The physical-current equation gives

$$\frac{dx}{dy} = \sqrt{1-bx^2}, x(0) = 0, x(y) = \frac{\sin(\sqrt{b}y)}{\sqrt{b}}. \quad (84)$$

At a source surface, the source law requires $x(y_R) = y_R$. The displayed alternative instead has $x < y$ for the relevant positive compactness values. It cannot retain both the same current law and the same continuous source-surface correspondence. If one keeps $x = y$ instead, its current is $r^2 J = GM/\sqrt{H}$, which violates the specified current balance. The selection therefore uses the combined field and source premises, before any orbit is compared.

Appendix B Following the alternative orbital route

To isolate what normalization does, hold the same exterior profile $A = 1 - R_s/r$ but choose the flat-radial control $\zeta^2 = 1$. This control shares the held readout Q_* with the selected solution. It does not satisfy all the field-selection premises of Sections 3 and 4.

The exact auxiliary equation from Section 8 becomes

$$r_{\sigma\sigma} - r \varphi_\sigma^2 = -\frac{GM}{(r - R_s)^2}, r^2 \varphi_\sigma = \ell_\sigma. \quad (85)$$

$$w'' + w = \frac{GM}{\ell_\sigma^2 (1 - R_s w)^2}. \quad (86)$$

This is exactly the spatial orbit of the earlier shifted-potential calculation, provided its angular momentum is identified as $\ell_\sigma = j/\varepsilon$ and its time is interpreted as σ . The match holds beyond the leading perihelion coefficient. Calling that auxiliary force a force in reference time T would be a different dynamical prescription.

For a direct comparison, choose the same two areal turning radii in both branches. Their conserved j and ε follow from the same turning-point equations because the radial velocity vanishes there. The total-angle integrand differs through the radial response:

$$\frac{d\varphi}{d\xi} = \frac{\sqrt{H(A(\xi))}}{\sqrt{1 - 2(3 + e \cos \xi)/P}}, H = A \zeta^2. \quad (87)$$

The selected branch has $H = 1$; the flat-radial control has $H = A$. At $e = 0.2$, independent orbit integration and quadrature give:

P	Radial response	Advance from quadrature rad	Advance from orbit integration rad
100	$\zeta^2 = 1$	0.1322932660	0.1322932660
100	$\zeta^2 = 1/A$	0.1974475160	0.1974475160
1000	$\zeta^2 = 1$	0.0126295320	0.0126295320
1000	$\zeta^2 = 1/A$	0.0189349966	0.0189349966

The agreement of two formulations within the flat-radial branch is exact, while the two radial branches give different accumulated azimuth. The difference is not removed by a consistent change of clock parameter. It is why the field-response selection must precede the orbital interpretation of Q_* .

The weak field contribution of radial response

The leading difference can also be seen directly in the action. For this comparison only, write $\zeta^2 = A^{-\lambda}$, with $\lambda=0$ for the control and $\lambda=1$ for the selected response. Let $U_N = GM/r$ and $v^2 = v_r^2 + v_t^2$, with these coordinate velocities taken per T . Expansion gives

$$\frac{L_p}{m_p} = -c^2 + \frac{v^2}{2} + U_N + \frac{L_1}{c^2} + O(c^{-4}), \quad (88)$$

$$L_1 = \frac{v^4}{8} + \frac{U_N v^2}{2} + \lambda U_N v_r^2 + \frac{U_N^2}{2}. \quad (89)$$

To see the advance without inserting its coefficient, expand the exact general orbit equation at fixed $p_0 = j^2/(GM)$. With $\ell_g = GM/c^2$, the result is

$$w'' + w = \frac{1}{p_0} + \ell_g \left[(1+2\lambda)w^2 + \frac{2(1-\lambda)w}{p_0} + (1-\lambda)(w')^2 \right] + O\left(\frac{\ell_g^2}{p_0^3}\right). \quad (90)$$

Substitute $w_0 = (1 + e \cos \varphi)/p_0$ into the first-order terms. The resonant coefficient is $2(2+\lambda)e\ell_g/p_0^2$. The same resonance calculation used in Section 10 therefore gives

$$\Delta\varphi = \frac{2\pi(2+\lambda)GM}{p c^2} + O\left(\frac{G^2 M^2}{p^2 c^4}\right). \quad (91)$$

Specified radial response	Leading coefficient	Mercury arcsec per century
$\zeta^2 = 1$	$4\pi GM/(p c^2)$	28.6538
$\zeta^2 = 1/A$	$6\pi GM/(p c^2)$	42.9807

The radial term in equation (89) vanishes instantaneously at periapsis because $v_r = 0$, yet contributes to the finite orbit as the planet approaches and leaves it. A calculation confined to the peak readout cannot capture that contribution. Splitting the total advance into time and spatial pieces depends on the chosen representation; the accumulated apsidal angle is the comparison quantity.

Responses differing only at higher weak-field order can preserve the leading 6π coefficient while changing exact orbits. Agreement with the leading Mercury number alone is therefore weaker than the source-family selection proved in Section 4.

Appendix C Normalization over the perihelion passage

The normalized readout is greatest at perihelion, but its accumulated effect also depends on the time spent there. For a held clock and a moving clock, respectively, the leading rate changes are

$$\frac{Q_*}{Q_0} - 1 = \frac{GM}{rc^2} + O(c^{-4}), \quad (92)$$

$$\frac{Q_{\text{rec}}}{Q_0} - 1 = \frac{1}{c^2} \left(\frac{GM}{r} + \frac{v^2}{2} \right) + O(c^{-4}). \quad (93)$$

Use the leading Kepler orbit, which is sufficient for these first-order clock effects. If E denotes eccentric anomaly and $n = \sqrt{GM/a^3}$, then

$$r = a(1 - e \cos E), dT = \frac{1 - e \cos E}{n} dE. \quad (94)$$

The radius dependence cancels from the leading gravitational clock-deficit integral:

$$\Delta T_h \equiv \int (1 - \alpha) dT \simeq \frac{GM}{ac^2 n} \Delta E. \quad (95)$$

Thus the deficit is uniform per eccentric-anomaly increment at this order, although the instantaneous rate change is larger near perihelion. For a full orbit of period $P_T = 2\pi/n$, the gravitational and moving-clock deficits are

$$\Delta T_h \simeq P_T \frac{GM}{ac^2}, \Delta T_{\text{moving}} \simeq \frac{3}{2} P_T \frac{GM}{ac^2}. \quad (96)$$

The second expression follows from $v^2 = GM(2/r - 1/a)$. With the Mercury inputs of Section 11, the self-consistent Kepler period is 87.96947 days. The resulting full-orbit deficits are 0.193806 seconds and 0.290710 seconds. These are calculated clock comparisons in the isolated solar model, not observations of a clock on Mercury.

A sum of held-clock increments along successive orbital locations is not the reading of one clock held at a fixed radius. The moving-clock quantity does follow the model orbit. Mercury's own gravitational field, other bodies and corrections beyond the stated approximation are not included in these numerical comparisons.

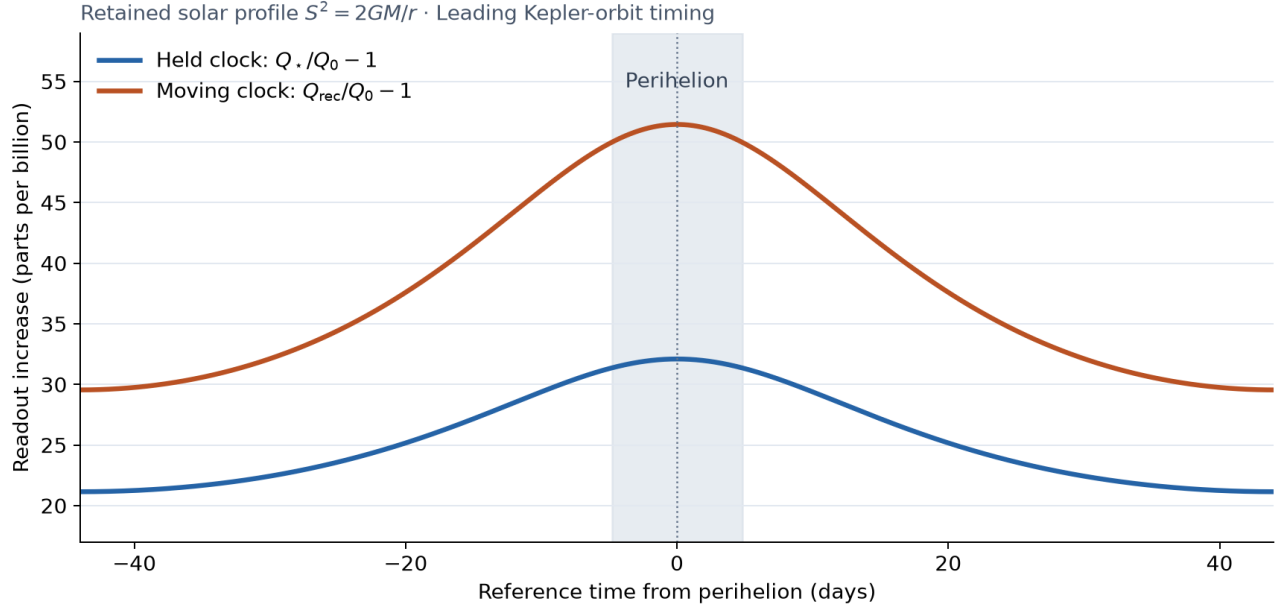
The finite interval around perihelion

Define a finite passage by true anomaly between -30° and $+30^\circ$. Its conversion to eccentric anomaly is

$$\tan \frac{E}{2} = \sqrt{\frac{1-e}{1+e}} \tan \frac{\varphi}{2}. \quad (97)$$

The interval lasts 9.6021 days. The gravitational clock deficit is 26.4235 milliseconds; including orbital motion gives 42.2698 milliseconds. It contains 10.9153 percent of the orbital time and 13.6340 percent of the leading gravitational clock deficit.

Mercury: receiver normalization along one orbit



Shaded interval: $\pm 30^\circ$ of true anomaly, lasting 9.60 days.
 Gravitational clock deficit: 26.4 ms; including orbital motion: 42.3 ms.
 The curves compare clock units for the same specified readout.

Figure 1. Receiving-clock normalization through one leading Mercury orbit in the retained solar field. The shaded region is the finite passage defined above. The held-clock increase is 32.10 parts per billion at perihelion and 21.15 at aphelion; the moving-clock increase is 51.45 and 29.55 parts per billion, respectively.

The time normalization conserves the same specified readout point by point:

$$Q_* d\tau_h = Q_0 dT, \int Q_* d\tau_h = \int Q_0 dT. \quad (98)$$

This remains true when $Q_0(r)$ varies along the orbit. The larger rate per receiving-clock second does not provide an additional source quantity to add to the force. Its dynamical role is already carried by the normalized potential and time conversion derived in Section 8.

Appendix D The free fall representation after selection

The main calculation used the capacity norm, physical work and source consistency. A coordinate representation provides an additional check. Before selecting H , multiplication of the invariant by dT^2 gives

$$c^2 d\tau^2 = A c^2 dT^2 - \frac{H}{A} dr^2 - r^2 d\Omega^2. \quad (99)$$

Write $h = \sqrt{H}$. For inward marginal-energy radial motion, $\varepsilon = 1$ and $j = 0$ give

$$\frac{dT}{d\tau} = \frac{1}{A}, \frac{dr}{d\tau} = -\frac{S}{h}. \quad (100)$$

Define a time coordinate by

$$dt_f = dT + \frac{Sh}{Ac^2} dr. \quad (101)$$

The coefficient depends only on radius, so this is locally an integrable change of stationary synchronization. Along the stated free-fall trajectories, $dt_f/d\tau = 1/A - S^2/(Ac^2) = 1$. Substitution into equation (99) gives

$$c^2 d\tau^2 = c^2 dt_f^2 - (h dr + S dt_f)^2 - r^2 d\Omega^2. \quad (102)$$

This identity holds for every positive H before selection. It identifies $d\chi = h dr = \alpha d\ell$ as the radial increment on these free-fall slices. The work-current defined in Section 3 is accordingly

$$J = -\frac{1}{2} \frac{dS^2}{d\chi} = \frac{c^2}{2h} \frac{dA}{dr}. \quad (103)$$

After the source-family result sets $H = 1$, equation (102) is the usual ingoing Painlevé–Gullstrand form of the selected exterior [7]. Its appearance is a consequence and coordinate crosscheck. No inference that S is a literal material outflow is required by this transformation.

The spatial volume element on this unit-energy free-fall slicing is $4\pi r^2 h dr$. It equals the spherical reference-shell volume after selection. That equality has not been used as an additional assumption to obtain $H = 1$, and it is not asserted for all observer slicings.

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The numerical checks accompanying this derivation include exact algebraic identities, direct orbit integrations, independent angular quadratures and the finite clock integrals in Appendix C. They validate the displayed calculations under their stated premises. They do not constitute additional astronomical measurements.